Half-degree irreducibles over $\mathbb{F}_2$: what is proved and what remains open

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Status. This note does not prove the Kaser–Lemire conjecture. The estimate shown in red in Figure 1 is still open.

Abstract

Kaser and Lemire observed that Barrett reduction over $\mathbb{F}_2$ is especially cheap when an irreducible polynomial f of degree n satisfies $\deg(f - x^n) \le \lfloor n/2 \rfloor$. They conjectured that such a polynomial exists in every degree. We translate their condition into a question about one ray class and then count that class with the polynomial von Mangoldt function. Most of the resulting character sum can now be bounded. The part we cannot bound is a signed sum over characters of high 2-power order. We state the needed bound and prove that it would finish the argument. Separate Rabin tests cover every degree through 400.

1 The argument

Put $\ell = \lceil n/2 \rceil - 1$. Figure 1 shows what is proved and what is not.

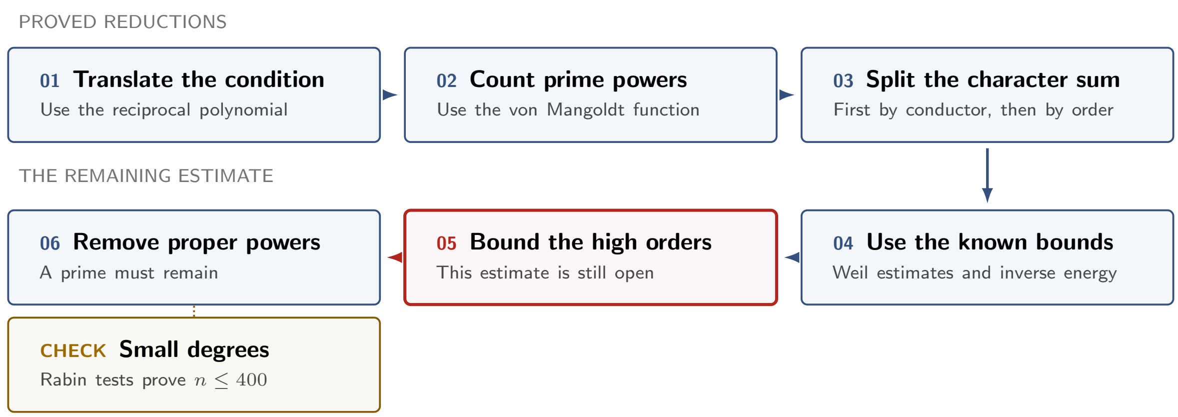

Figure 1: Steps 1–4 and 6 are proved. Step 5 is open. The gold box records the separate computation for $n \le 400$.

2 What we can prove

1. Translate the condition (PROVED). For $j \ge 0$ set

$$\mathcal{E}_j = (1 + x\mathbb{F}_2[x])/(x^{j+1}), \qquad |\mathcal{E}_j| = 2^j,$$

and, for monic F, let $\langle F \rangle_j = x^{\deg F} F(x^{-1}) \pmod{x^{j+1}}$. Then $\deg(f - x^n) \le \lfloor n/2 \rfloor$ if and only if $\langle f \rangle_\ell = 1$. So the conjecture asks whether this ray class contains a prime of degree n.

2. Count prime powers (PROVED). Define

$$N_j(b) = \sum_{\substack{F \text{ monic, } \deg F = n \\ \langle F \rangle_j = b}} \Lambda(F).$$

Logarithmic differentiation gives this weighted count exactly. At the odd endpoint $n = 2\ell + 1$, $N_\ell(1) = 1 + nI_n(1)$. At the even endpoint $n = 2\ell + 2$, odd proper powers do not occur in the identity class. A direct count bounds the squares and the other even powers. In either case, it is enough to prove

$$N_\ell(1) > 2^{n-\ell} - 2^\ell. \tag{1}$$

3. Split the count by conductor (PROVED). Each map $\mathcal{E}_j \to \mathcal{E}_{j-1}$ has two-element fibres. If $H_j(b)$ is the difference of the two child populations above b, repeated binary splitting gives

$$2^\ell N_\ell(e) = 2^n + \sum_{j=1}^{\ell} \varepsilon_j(e) 2^{j-1} H_j(b_j(e)), \qquad \varepsilon_j(e) \in \{\pm 1\}. \tag{2}$$

For primitive characters X_j of conductor j,

$$H_j(b) = 2^{1-j} \sum_{\chi \in X_j} \overline{\chi(b)} S_n(\chi), \quad |H_j(b)| \le (j-1) 2^{\lceil n/2 \rceil},$$

The function-field Riemann hypothesis gives the inequality on the right. It is strong enough for the small conductors. If we use it separately at every conductor, however, the final bound is too large by a factor of order ℓ.

4. Prove the bilinear bound (PROVED, but not enough). The needed inverse-energy bound holds in characteristic two, including the case where products wrap modulo x^r. The proof groups collisions by their x-adic valuation. A Padé approximation turns each group into a problem about factoring polynomials of bounded degree. Lift counts and divisor bounds then finish the estimate. This proves the bilinear input used in Bagshaw's Type-I/II argument. It does not finish our problem, because applying the bound one term at a time destroys the signs in (2).

5. Add the large conductors before taking absolute values (PROVED). Let $c = \lceil \log_2 \ell \rceil$, $a = \ell - c - 1$, and

$$W_{\ell,n} = 2^{\lceil n/2 \rceil} \sum_{1 \le j < a} (j-1) 2^{j-1}, \quad B_{\ell,n} = 2^{2\ell} - W_{\ell,n} > 0.$$

For the identity class, these terms telescope:

$$C_{\ell,n} := \sum_{j=a}^{\ell} 2^{j-1} H_j(1) = 2^\ell N_\ell(1) - 2^{a-1} N_{a-1}(1). \tag{3}$$

It would now be enough to prove

$$C_{\ell,n} > -B_{\ell,n} \tag{REL}$$

The smaller conductors contribute at least $-W$. With (REL), this gives $2^\ell N_\ell(1) - 2^n > -W - B = -2^{2\ell}$, which is (1).

6. Split the remaining characters by order (PROVED). For $s \ge 0$, write

$$\mathcal{H}_{j,s} = \{\chi : \chi^{2^s} = 1\}, \quad h_{j,s} = 2^{j - \lfloor j/2^s \rfloor}, \quad P_{j,s} = \sum_{g \in 2^s \mathcal{E}_j} N_j(g).$$

Finite-group orthogonality gives the sum over characters of conductor j and exact order 2^s:

$$T_{j,s} = h_{j,s} P_{j,s} - h_{j,s-1} P_{j,s-1} - h_{j-1,s} P_{j-1,s} + h_{j-1,s-1} P_{j-1,s-1}.$$

Let Q be the largest power of two with $3Q \lceil \log_2 \ell \rceil \le \ell$. The Weil bound handles every order at most Q. This leaves $O(\log\log \ell)$ high orders at each conductor. When $\ell = 200$, it handles 20 of the 67 conductor–order pairs and leaves 47.

3 What remains open

7. Bound the high orders (OPEN). Suppose $a \le j \le \ell$, $2^s > Q$, and the corresponding set of characters is nonempty. The following bound would be enough:

$$4\ell \, |T_{j,s}(n)| \le \# X_{j,s} (j-1) 2^{\lceil n/2 \rceil}. \tag{HWO}$$

We can also state the same bound without characters. Put $\Delta_{j,s} = 2P_{j,s} - P_{j-1,s}$, $d_s = \lfloor (j-1)/2^s \rfloor$, and $R_{j,s} = 2^{d_{s-1} - d_s}$. Then (HWO) is equivalent to

$$4\ell \, |R_{j,s} \Delta_{j,s} - \Delta_{j,s-1}| \le (R_{j,s} - 1)(j-1) 2^{\lceil n/2 \rceil}. \tag{NSD}$$

For these orders, $2^s > \ell/(3c)$. The group $2^s \mathcal{E}_j$ therefore has fewer than $8\ell^3$ elements. That fact narrows the problem, but it does not solve it. At the largest order the group is $\{1\}$, so (NSD) still contains the original identity-class term.

If we prove this bound, the rest follows:

$$\boxed{\text{HWO (or another proof of REL)}} \Longrightarrow \text{REL} \Longrightarrow (1) \Longrightarrow \text{prime in the identity class.}$$

Step 2 removes the proper powers. Checked Rabin tests cover $n \le 400$ (FINITE). A proof of (HWO) for $\ell \ge 200$ and $n \in \{2\ell+1, 2\ell+2\}$ would prove the conjecture.

Why the usual bounds fail. Taking absolute values for each character or conductor loses too much. Bounds that use only the Fourier support give no saving at all. Cauchy's inequality, the tested bent and Kerdock models, and simple Heisenberg lifts also fail. Known moment theorems over large finite fields are not uniform in this fixed-field, growing-conductor setting. We need a bound that keeps the high-order character sums together. Another table of small cases will not provide it.

References

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